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Enriching continuous Lagrange finite element approximation spaces using neural networks

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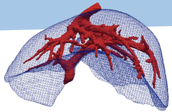
PhD student seminar

Scientific context



Context : Create real-time digital twins of an organ (e.g. liver).

Objective : Develop an hybrid finite element / neural network method.

finite element: accurate
neural network: quick + parameterized

Parametric elliptic convection/diffusion PDE : For one or several $\boldsymbol{\mu} \in \mathcal{M}$, find $u : \Omega \to \mathbb{R}$ such that

$$\mathcal{L}\big(u\,;\boldsymbol{x},\boldsymbol{\mu}\big) = f(\boldsymbol{x},\boldsymbol{\mu}), \qquad (\mathcal{P})$$

where $\mathcal{L}$ is the parametric differential operator defined by

$$\mathcal{L}(\cdot;\boldsymbol{x},\boldsymbol{\mu}) : u \mapsto R(\boldsymbol{x},\boldsymbol{\mu})u + C(\boldsymbol{\mu})\cdot\nabla u - \frac{1}{\mathrm{Pe}}\nabla\cdot\big(D(\boldsymbol{x},\boldsymbol{\mu})\nabla u\big),$$

and some Dirichlet, Neumann or Robin BC (which can also depend on $\boldsymbol{\mu}$).

Pipeline of the Enriched FEM

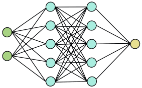
OFFLINE : PINN training

Inputs

$\boldsymbol{x} \in \Omega$: space coordinates

$\boldsymbol{\mu} \in \mathcal{M}$: parameter

$\longrightarrow$



Parametric PINN

$\longrightarrow$

Output

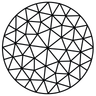
$u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$: prediction of the PDE solution $u(\boldsymbol{x}, \boldsymbol{\mu})$

ONLINE (fixed $\boldsymbol{\mu}$) : PINN evaluation + Enriched FEM resolution

PINN evaluation

$u_\theta(\cdot, \boldsymbol{\mu})$: prediction for the given parameter μ

$\longrightarrow$



FEM solver enriched by u_θ

$\longrightarrow$

Output

enriched FEM solution (depending on the mesh size h)

Complete ONLINE process : quick + accurate

Classical approaches

Physics-Informed Neural Networks

Standard PINNs[1] **(Weak BC) :** Find the optimal weights $\theta^\star$, such that

$$\theta^\star = \underset{\theta}{\operatorname{argmin}} \big(\omega_r J_r(\theta) + \omega_b J_b(\theta)\big), \qquad (\mathcal{P}_\theta)$$

with

residual loss $$\boxed{J_r(\theta) = \int_{\mathcal{M}} \int_\Omega \left| \mathcal{L}\big(u_\theta(\boldsymbol{x},\boldsymbol{\mu}); \boldsymbol{x},\boldsymbol{\mu}\big) - f(\boldsymbol{x},\boldsymbol{\mu}) \right|^2 d\boldsymbol{x} d\boldsymbol{\mu},}$$

boundary loss $$\boxed{J_b(\theta) = \int_{\mathcal{M}} \int_{\partial\Omega} \left| u_\theta(\boldsymbol{x},\boldsymbol{\mu}) - g(\boldsymbol{x},\boldsymbol{\mu}) \right|^2 d\boldsymbol{x} d\boldsymbol{\mu},}$$

where u_θ is a neural network, $g = 0$ is the Dirichlet BC.

In $(\mathcal{P}_\theta)$, ω_r and ω_b are some weights.

Monte-Carlo method : Discretize the cost functions by random process.

[1] [Raissi et al., 2019]

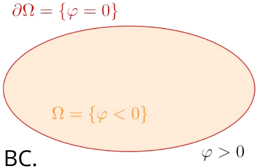
Physics-Informed Neural Networks

Improved PINNs[1] (Strong BC) : Find the optimal weights $\theta^\star$ such that

$$\theta^\star = \underset{\theta}{\operatorname{argmin}} \left(\omega_r J_r(\theta) + \cancel{\omega_b J_b(\theta)} \right),$$

with $\omega_r = 1$ and

residual loss
$$\boxed{J_r(\theta) = \int_{\mathcal{M}} \int_\Omega \left| \mathcal{L}\big(u_\theta(\boldsymbol{x},\boldsymbol{\mu});\boldsymbol{x},\boldsymbol{\mu}\big) - f(\boldsymbol{x},\boldsymbol{\mu}) \right|^2 d\boldsymbol{x} d\boldsymbol{\mu},}$$



where u_θ is a neural network defined by

$$u_\theta(\boldsymbol{x},\boldsymbol{\mu}) = \varphi(\boldsymbol{x}) w_\theta(\boldsymbol{x},\boldsymbol{\mu}) + g(\boldsymbol{x},\boldsymbol{\mu}),$$

with φ a level-set function, w_θ a NN and $g = 0$ the Dirichlet BC.

Thus, the Dirichlet BC is imposed exactly in the PINN : $u_\theta = g$ on $\partial\Omega$.

[1] [Lagaris et al., 1998; Franck et al., 2024]

Finite Element Method[1]

Variational Problem :

$$\text{Find } u_h \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(u_h, v_h) = l(v_h), \qquad (\mathcal{P}_h)$$

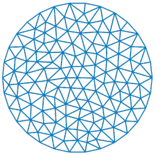
with h the characteristic mesh size, a and l the bilinear and linear forms given by

$$a(u_h, v_h) = \frac{1}{\text{Pe}} \int_\Omega D\nabla u_h \cdot \nabla v_h + \int_\Omega R\, u_h\, v_h + \int_\Omega v_h\, C \cdot \nabla u_h, \quad l(v_h) = \int_\Omega f\, v_h,$$

and V_h^0 the finite element space defined by

$$V_h^0 = \left\{ v_h \in C^0(\Omega),\ \forall K \in \mathcal{T}_h,\ v_h|_K \in \mathbb{P}_k, v_h|_{\partial\Omega} = 0 \right\},$$

where $\mathbb{P}_k$ is the space of polynomials of degree at most k.



$\mathcal{T}_h = \{K_1, \ldots, K_{N_e}\}$
(N_e : number of elements)

Linear system : Let $(\phi_1, \ldots, \phi_{N_h})$ a basis of V_h^0.

Find $U \in \mathbb{R}^{N_h}$ such that $\qquad AU = b$

with

$$A = \big(a(\phi_i, \phi_j)\big)_{1 \le i,j \le N_h} \quad \text{and} \quad b = \big(l(\phi_j)\big)_{1 \le j \le N_h}.$$

[1] [Ern and Guermond, 2004]

Enriched finite element method using PINN

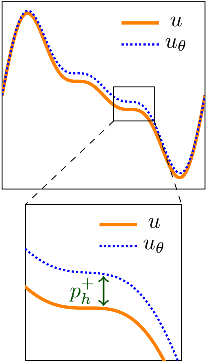
Additive approach

Variational Problem : Let $u_\theta \in H^{k+1}(\Omega) \cap H_0^1(\Omega)$ be a PINN prior.

$$\text{Find } p_h^+ \in V_h^0 \text{ such that, } \forall v_h \in V_h^0, a(p_h^+, v_h) = l(v_h) - a(u_\theta, v_h), \qquad (\mathcal{P}_h^+)$$

with the enriched trial space V_h^+ defined by

$$V_h^+ = \left\{ u_h^+ = u_\theta + p_h^+, \quad p_h^+ \in V_h^0 \right\}.$$



General Dirichlet BC : If $u = g$ on $\partial\Omega$, then

$$p_h^+ = g - u_\theta \quad \text{on } \partial\Omega,$$

with u_θ the PINN prior.

We hope that the modified problem will give the same results as the standard one on coarser meshes.

Convergence analysis

Theorem 1: Convergence analysis of the standard FEM [Ern and Guermond, 2004]

We denote $u_h \in V_h$ the solution of $(\mathcal{P}_h)$ with V_h the standard trial space. Then,

$$|u - u_h|_{H^1} \leqslant C_{H^1}\, h^k |u|_{H^{k+1}},$$

$$\|u - u_h\|_{L^2} \leqslant C_{L^2}\, h^{k+1} |u|_{H^{k+1}}.$$

Theorem 2: Convergence analysis of the enriched FEM [Lecourtier et al., 2025]

We denote $u_h^+ \in V_h^+$ the solution of $(\mathcal{P}_h^+)$ with V_h^+ the enriched trial space. Then,

$$|u - u_h^+|_{H^1} \leqslant \boxed{\frac{|u-u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}}} \left(C_{H^1}\, h^k |u|_{H^{k+1}} \right),$$

$$\|u - u_h^+\|_{L^2} \leqslant \boxed{\frac{|u-u_\theta|_{H^{k+1}}}{|u|_{H^{k+1}}}} \left(C_{L^2}\, h^{k+1} |u|_{H^{k+1}} \right).$$

Gains of the additive approach.

Numerical results

- 2D Poisson problem on Square - Dirichlet BC
- 3D Poisson problem on Cube - Dirichlet BC

Numerical results

2D Poisson problem on Square - Dirichlet BC
3D Poisson problem on Cube - Dirichlet BC

Problem considered

Problem statement: Consider the Poisson problem in 2D with Dirichlet BC:

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \times \mathcal{M}, \\ u = 0, & \text{on } \partial\Omega \times \mathcal{M}, \end{cases}$$

with $\Omega = [-0.5\pi, 0.5\pi]^2$ and $\mathcal{M} = [-0.5, 0.5]^2$ ($p = 2$ parameters).

Analytical solution :

$$u(\boldsymbol{x}, \boldsymbol{\mu}) = \exp\left(-\frac{(x-\mu_1)^2 + (y-\mu_2)^2}{2}\right) \sin(2x)\sin(2y).$$

PINN training:

MLP 5 layers (sine - 40, 60, 60, 60, 40); LBFGs optimizer (5000 epochs - 6000 col. pts).

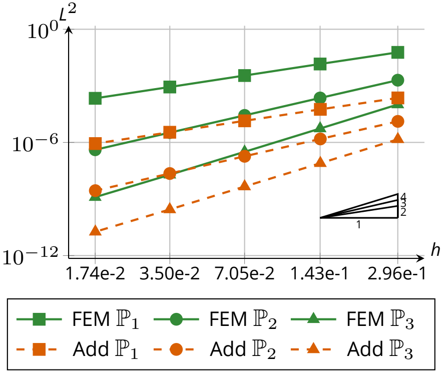
Imposing the Dirichlet BC exactly in the PINN with the levelset φ defined by

$$\varphi(\boldsymbol{x}) = (x + 0.5\pi)(x - 0.5\pi)(y + 0.5\pi)(y - 0.5\pi).$$

Online phase - 1 set of parameters

$$\boldsymbol{\mu}^{(1)} = (0.05, 0.22)$$

Error estimates in L^2 norm:



Online phase - 1 set of parameters

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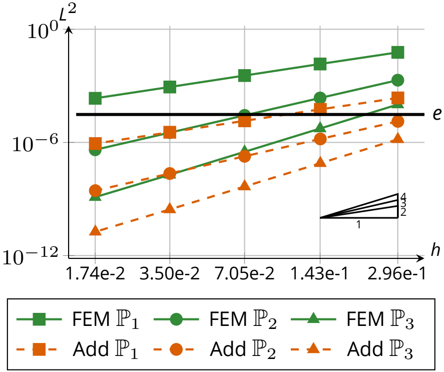


N_{dofs} required to reach the same error e:

		N_{dofs}	
k	**e**	**FEM**	**Add**
1	$1 \cdot 10^{-3}$	14 161	64
	$1 \cdot 10^{-4}$	143 641	576
2	$1 \cdot 10^{-4}$	6 889	225
	$1 \cdot 10^{-5}$	31 329	1 089
3	$1 \cdot 10^{-5}$	6 724	784
	$1 \cdot 10^{-6}$	20 164	2 704

Parametric framework

Considering the set of $n_p = 50$ parameter instances

$$\mathcal{S} = \left\{ \boldsymbol{\mu}^{(1)}, \dots, \boldsymbol{\mu}^{(n_p)} \right\}.$$

Gains achieved: $\|u - u_h\|_{L^2} / \|u - u_h^+\|_{L^2}$

	Gains in L^2 rel error of our method w.r.t. FEM		
k	**min**	**max**	**mean**
1	134.32	377.36	269.39
2	67.02	164.65	134.85
3	39.52	72.65	61.55

Cartesian mesh : N^2 nodes with $N = 20$.

Numerical results

2D Poisson problem on Square - Dirichlet BC

3D Poisson problem on Cube - Dirichlet BC

Problem considered

Problem statement: Consider the Poisson problem in 3D with Dirichlet BC:

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \times \mathcal{M}, \\ u = 0, & \text{on } \partial\Omega \times \mathcal{M}, \end{cases}$$

with $\Omega = [-0.5\pi, 0.5\pi]^3$ and $\mathcal{M} = [-0.5, 0.5]^3$ ($p = 3$ parameters).

Analytical solution :

$$u(\boldsymbol{x}, \boldsymbol{\mu}) = \exp\left(-\frac{(x-\mu_1)^2 + (y-\mu_2)^2 + (z-\mu_3)^2}{2}\right) \sin(2x)\sin(2y)\sin(2z).$$

PINN training:

MLP 5 layers (tanh - 40, 60, 60, 60, 40); LBFGs optimizer (5000 epochs - 40000 col. pts).

Imposing the Dirichlet BC exactly in the PINN with the levelset φ defined by

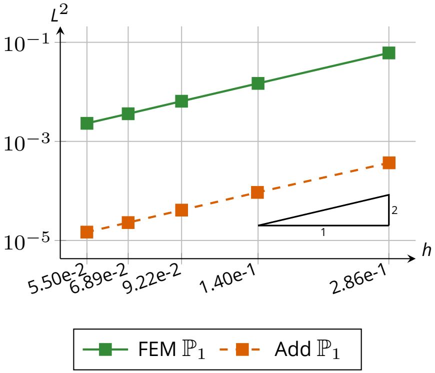
$$\varphi(\boldsymbol{x}) = (x+0.5\pi)(x-0.5\pi)(y+0.5\pi)(y-0.5\pi)(z+0.5\pi)(z-0.5\pi).$$

Training time : 12 minutes on NVIDIA RTX 2000 (8GB VRAM).

Online phase - 1 set of parameters I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

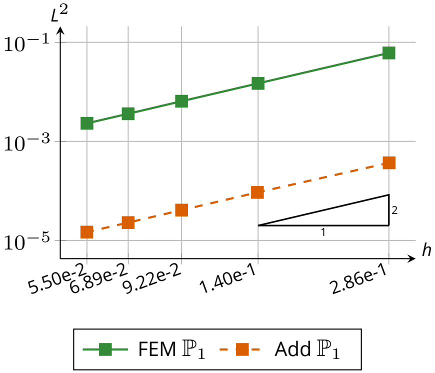
Error estimates in L^2 norm:



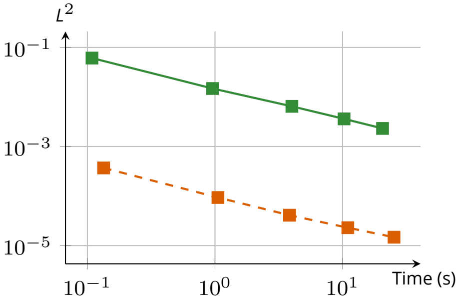
Online phase - 1 set of parameters I

$$\boldsymbol{\mu}^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:



L^2 error w.r.t. execution time*:



* mesh + assembly + solve

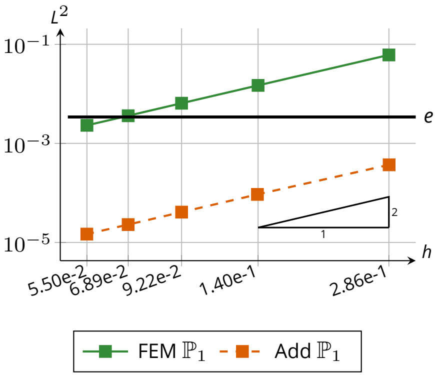
+ prior derivatives evaluation

~~+ training time~~

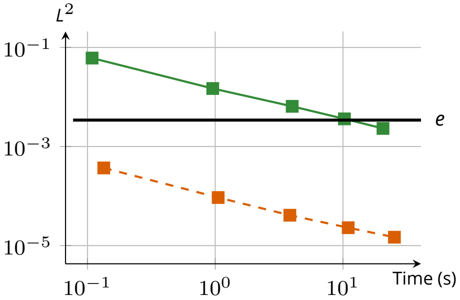
Online phase - 1 set of parameters I

$$\mu^{(1)} = (0.05, 0.22, 0.1)$$

Error estimates in L^2 norm:



L^2 error w.r.t. execution time*:



* mesh + assembly + solve

+ prior derivatives evaluation

~~+ training time~~

Online phase - 1 set of parameters II

$$\boldsymbol{\mu}^{(1)} = (0.05, 0.22, 0.1)$$

N_{dofs} and computation time required to reach the same error e:

	N		N_{dofs}		Computation time*	
e	**FEM**	**Add**	**FEM**	**Add**	**FEM**	**Add**
$1 \cdot 10^{-3}$	152	12	$3.51 \cdot 10^{6}$	$1.73 \cdot 10^{3}$	$(7.65 \cdot 10^{1})$	$5.2 \cdot 10^{-2}$
$1 \cdot 10^{-4}$	484	39	$1.13 \cdot 10^{8}$	$5.93 \cdot 10^{4}$	$(2.8 \cdot 10^{3})$	$9.26 \cdot 10^{-1}$
$1 \cdot 10^{-5}$	1 539	122	$3.65 \cdot 10^{9}$	$1.82 \cdot 10^{6}$	$(1.02 \cdot 10^{5})$	$(5.26 \cdot 10^{1})$

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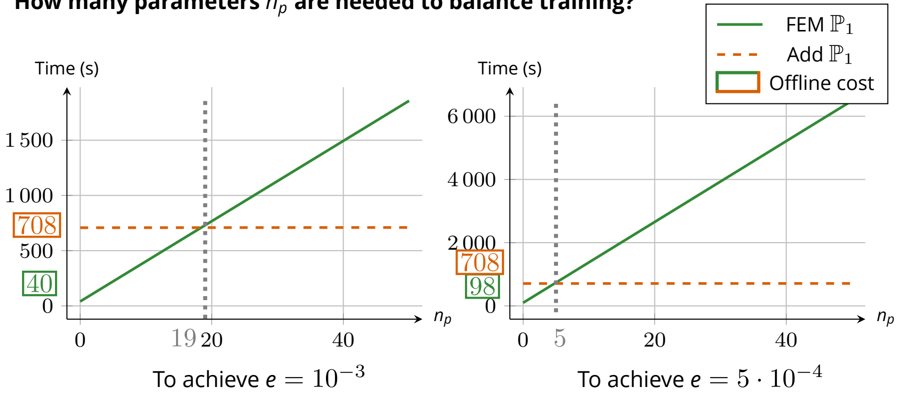
Cartesian mesh : N^3 nodes; $\mathbb{P}_1$ elements.

* The results in brackets are extrapolations.

Parametric framework

How many parameters n_p are needed to balance training?



Online: assembly + solve + prior derivatives evaluation

Offline: mesh + training time

Conclusion

TODO

Preprint (linear)

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Preprint (linear)

Appendix 1 : Sec1

A1 – Appendix 1

TODO